

TNA Core Language v2

A Typed Meta-Structural Calculus for Domain-Limited Systems

0. Meta-Level Definition

TNA operates over **typed configuration spaces** and defines:

- admissibility
- boundary behavior
- domain evolution

It is **not embedded in spacetime**.

It defines the **conditions under which spacetime-like models are valid**.

1. Type System

1.1 Base Types

Domain	::= Ω
Boundary	::= $\partial\Omega$
Config	::= $x \in \Omega$
Transition	::= T
Function	::= $f: \Omega \rightarrow \mathbb{R}$
Operator	::= $Op(\Omega) \rightarrow \Omega'$

1.2 Extended Types

Entropy	::= S_S ∈ ℝ ⁺
Expansion	::= H_S ∈ ℝ
Reliability	::= R: Ω → [0,1]
Pressure	::= F_B ∈ ℝ
QuestionRate	::= F_Q ∈ ℝ
Information	::= I(Ω)

1.3 Typed System Object

$$\Sigma := (\Omega, \partial\Omega, T, R, S_s, H_s, \mathcal{F}_B)$$

2. Admissibility Logic

2.1 Predicate

$$\mathcal{A}(x, \Omega) = \begin{cases} 1 & x \in \Omega \\ 0 & x \notin \Omega \end{cases}$$

2.2 Boundary Condition

$$x \in \partial\Omega \iff \lim_{\epsilon \rightarrow 0} \mathcal{A}(x + \epsilon) \text{ is undefined or non-unique}$$

2.3 Invalid State

$$x \notin \Omega \Rightarrow \text{requires } T : \Omega \rightarrow \Omega'$$

3. Structural Dynamics

3.1 Entropy

$$S_s = \log |\Omega|$$

3.2 Expansion

$$H_s = \frac{dS_s}{dt}$$

3.3 Reliability Field

$$R(x) = 1 - \delta(x, \partial\Omega)$$

where:

$$\delta(x, \partial\Omega) \rightarrow 1 \text{ as } x \rightarrow \partial\Omega$$

3.4 Evolution Equation (Typed)

$$\frac{D^2\Omega}{Dt^2} + \Gamma^i_{jk} \frac{D\Omega^j}{Dt} \frac{D\Omega^k}{Dt} = \mathcal{F}_B$$

Constraints:

- Defined in **configuration manifold**
 - D/Dt = structural derivative (not physical time)
-

4. Operators

4.1 Expansion Operator

$\text{Expand}(\Omega) \rightarrow \Omega^+$

Constraint:

$$|\Omega^+| > |\Omega|$$

4.2 Boundary Detection

$\text{Boundary}(x, \Omega) \rightarrow \text{Bool}$

True if:

$$R(x) < \epsilon$$

4.3 Transition Operator

$\text{Transition}(\Omega) \rightarrow \Omega'$

Triggered when:

- admissibility fails
 - extension required
-

4.4 Collapse Operator

$\text{collapse}(x, \Omega) \rightarrow \perp$

Meaning:

- invalid state

- requires reconfiguration
-

4.5 Projection Operator

$\text{Proj}(\Omega_a \rightarrow \Omega_b)$

Maps between domains with possible information loss.

5. Axioms (Typed Form)

A1 — Non-Null Domain

$\text{Empty} \notin \text{Domain}$

$$\emptyset \notin \Omega$$

A2 — Boundary Existence

$\forall \Omega: \partial \Omega \neq \emptyset$

A3 — Expansion Persistence

$\text{Expand}(\Omega) \Rightarrow \text{Expand}(\partial \Omega)$

A4 — Boundary Indeterminacy

$x \in \partial \Omega \Rightarrow \text{Extensions}(x) > 1$

A5 — Information Preservation

$$I(\Omega_{\text{old}}) \subseteq I(\Omega_{\text{new}})$$

6. Inference Rules

R1 — Boundary $\rightarrow$ Indeterminacy

$$x \in \partial\Omega \Rightarrow |\text{Extensions}(x)| > 1$$

R2 — Indeterminacy $\rightarrow$ Transition Pressure

$$|\text{Extensions}(x)| > 1 \Rightarrow \mathcal{F}_B > 0$$

R3 — Pressure $\rightarrow$ Expansion

$$\mathcal{F}_B > 0 \Rightarrow \text{Expand}(\Omega)$$

R4 — Expansion $\rightarrow$ New Boundary

$$\text{Expand}(\Omega) \Rightarrow \partial\Omega^+ \neq \partial\Omega$$

R5 — Cognitive Mapping

$$\mathcal{F}_Q \sim \mathcal{F}_B$$

7. Multi-Domain System

7.1 Domain Set

$$\{\Omega_1, \Omega_2, \dots, \Omega_n\}$$

7.2 Transition Graph

$$G = (V, E), \quad V = \Omega_i, \quad E = T_{ij}$$

7.3 Time Definition

$$t := \text{ordered path in } G$$

8. Structural Metrics

8.1 Boundary Density

$$\rho_B = \frac{|\partial\Omega|}{|\Omega|}$$

8.2 Instability Index

$$\mathcal{I} = \int_{\Omega} (1 - R(x)) dx$$

8.3 Transition Likelihood

$$P(T) \propto \mathcal{F}_B$$

9. Execution Schema (IA-Oriented)

Canonical Loop

```
while system exists:
    detect boundary states
    compute indeterminacy
    update F_B
    if F_B > threshold:
        Ω = Expand(Ω)
```

10. Canonical Compression

Minimal Law

$$\partial\Omega \Rightarrow \Delta \Rightarrow \mathcal{F}_B \Rightarrow \Omega^+$$

Natural Language Form

Boundary generates indeterminacy; indeterminacy generates pressure; pressure drives expansion.

11. Interpretation Layer (Strictly Optional)

DOMAIN	BOUNDARY EXPRESSION
Physics	singularities, quantum indeterminacy
Cognition	existential questioning
ML systems	generalization failure
Complexity	phase transitions